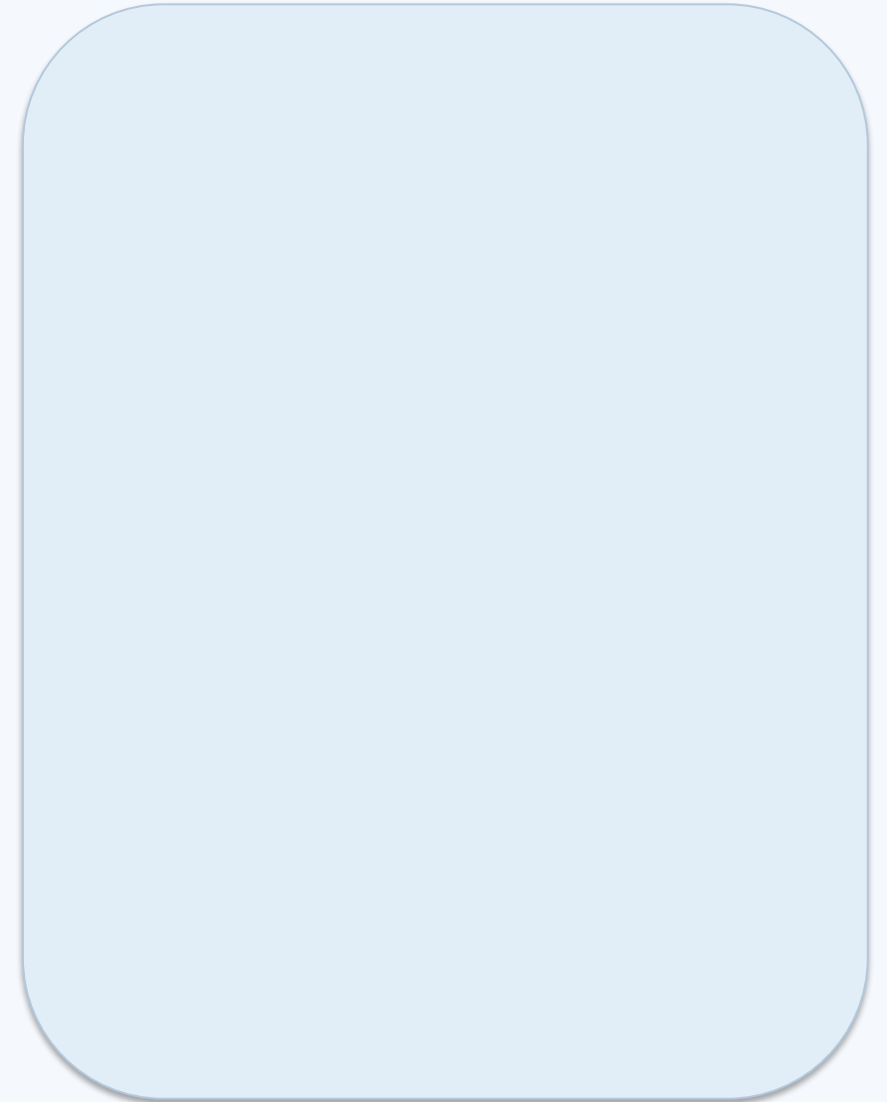


Electric Vehicles

Prepared by: Japanjot Singh

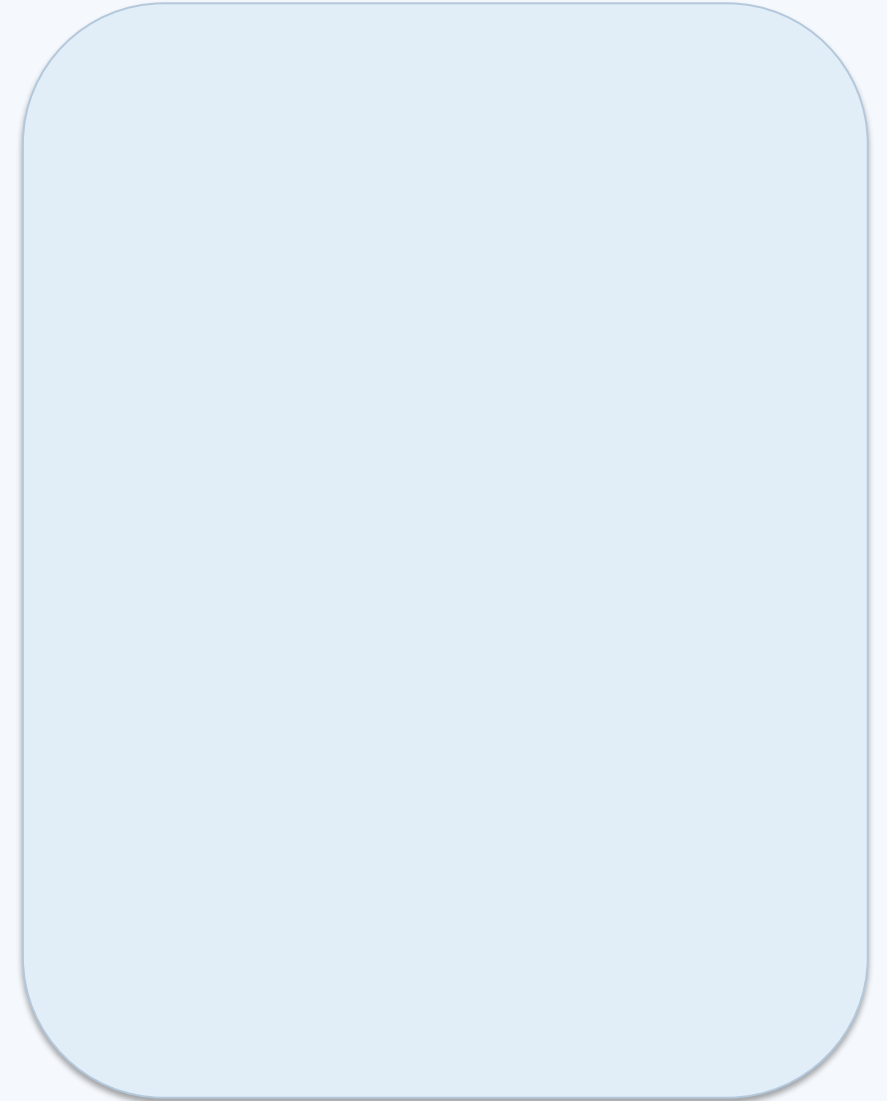
B.Tech Electrical Engineering

A modern overview of EV technology, benefits, challenges and future.



Agenda

- What are EVs?
- Types
- Components
- Benefits
- Challenges
- Market Growth
- Timeline
- Future
- Conclusion



What Are Electric Vehicles?

Electric Vehicles (EVs) use electric motors powered by rechargeable batteries instead of internal combustion engines.

Key Features:

- Zero tailpipe emissions
- High efficiency
- Low maintenance
- Regenerative braking

💡 Key Takeaways

- Important facts
- Presentation visuals
- Discussion points

Types of Electric Vehicles

- BEV – Fully electric
- HEV – Hybrid Electric Vehicle
- PHEV – Plug-in Hybrid
- FCEV – Hydrogen Fuel Cell

Each type balances cost, range and charging needs differently.

💡 Key Takeaways

- Important facts
- Presentation visuals
- Discussion points

Main Components

-  Battery Pack
-  Electric Motor
-  Power Electronics Controller
-  Charging Port
-  Regenerative Braking

These components work together to convert electrical energy into motion.

Key Takeaways

- Important facts
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- Discussion points

Benefits

- Environment friendly
- Lower running cost
- Reduced noise pollution
- High energy efficiency
- Government incentives
- Better driving experience

💡 Key Takeaways

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Challenges

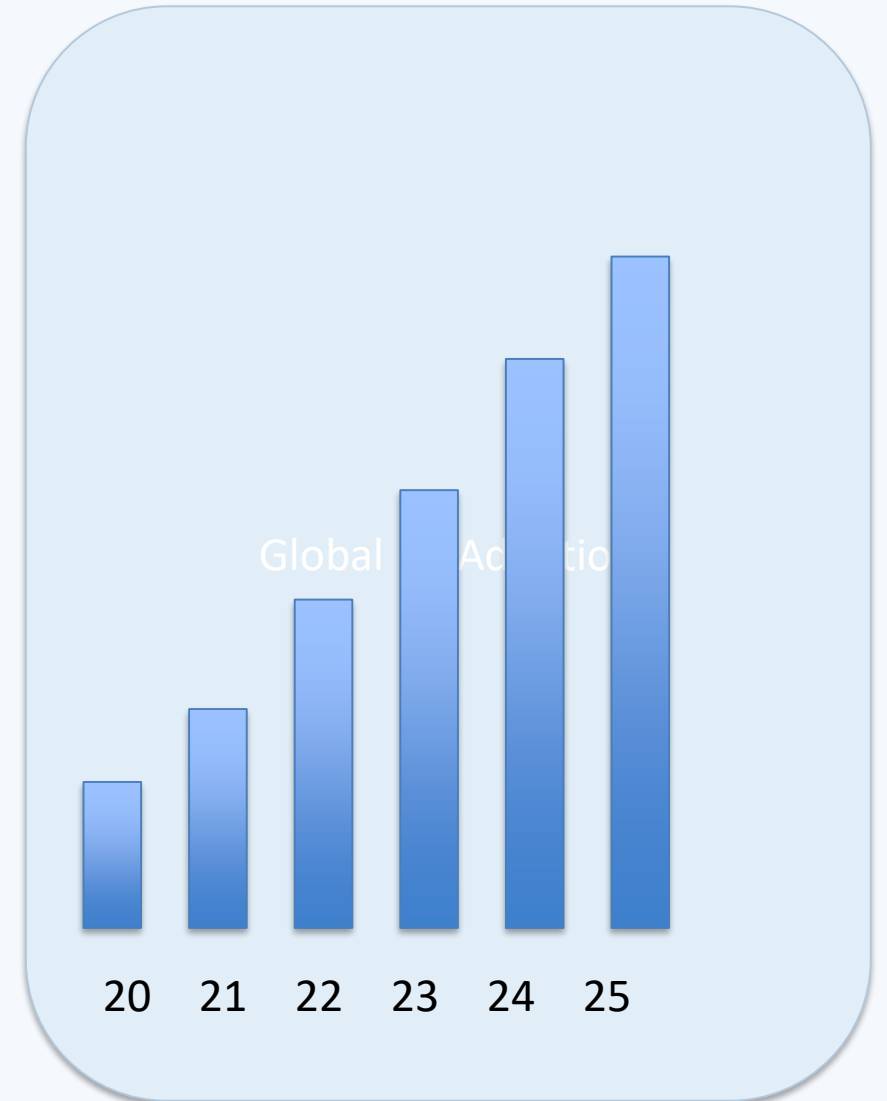
- Charging infrastructure
- Battery degradation
- Higher purchase price
- Charging time
- Raw material sourcing
- Recycling and disposal

💡 Key Takeaways

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Global EV Growth

Estimated increase in global EV adoption from 2020–2025.
See chart on right.



Evolution Timeline

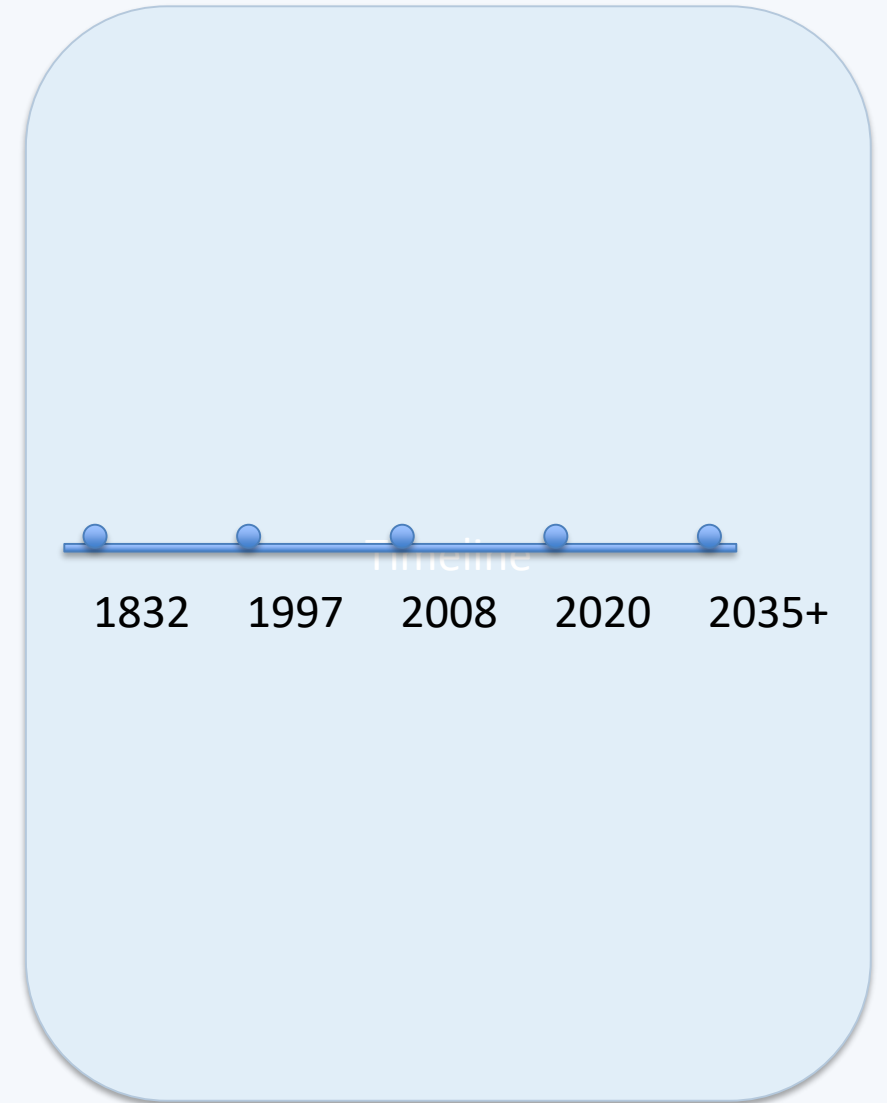
1832: First electric carriage

1997: Toyota Prius

2008: Tesla Roadster

2020: Mass-market expansion

2035+: Transition to cleaner mobility



Future & Conclusion

Future:

- Solid-state batteries
- Autonomous EVs
- Wireless charging
- Vehicle-to-grid technology

Conclusion:

Electric vehicles are transforming transportation toward a cleaner and more

💡 Key Takeaways

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